Anushka Singh

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Summary

ML Systems Engineer with experience optimizing deep learning workloads, GPU-accelerated AI systems, and memory-efficient LLM infrastructure. Skilled in CUDA, PyTorch, performance profiling, quantization, and systems-level optimization. MS CS graduate from Johns Hopkins University, focused on scalable AI infrastructure and efficient model execution.

Technical Skills

- **Programming**: Python, C/C++, CUDA
- **ML Systems & Inference**: vLLM, SGLang, PagedAttention, Continuous Batching, Speculative Decoding, KV Cache Compression, CUDA Memory Profiling, PyTorch Profiler
- **ML & AI**: PyTorch, Hugging Face Transformers, LLMs, Agentic AI, RAG, LangGraph, NLP, TensorFlow, Scikit-Learn
- **Tools & Platforms**: AWS, Docker, Git, Linux, Weights & Biases, REST APIs

Projects & Publications

Selective KV Cache Quantization for LLMs | *PyTorch, CUDA, Hugging Face, int8 Quantization* | ★ **Best Project Award**

- Designed a memory-efficient inference strategy for **LLaMA-3.1 8B** achieving **2× memory reduction** with near-zero perplexity loss and minimal ROUGE-L degradation via selective int8 quantization of the KV cache — enabling longer context on the same hardware budget.

Graph Neural Networks for Real-Time Fraud Detection | *Python, GNNs, GANs, AWS Lambda*

- Developed a **Generative Adversarial Graph Network** for financial transaction fraud detection, **reducing false positives by 40%** and achieving sub-second inference via AWS Lambda for real-time anomaly flagging.

Publication | Human Behavior and Emerging Technologies (Wiley), 2026

- **A Pilot Study of a Gamified CBT-Based Digital Approach for Body Image Distress and Appearance-Related Concerns.**

Publication | IEEE Confluence 2024

- **Impact of Colour Image and Skeleton Plotting on Sign Language Recognition Using CNN.**

Experience

Cornell Tech – Break Through Tech — New York, NY

AI Studio Coach — *Jan 2026 – May 2026*

- Coach 4+ interdisciplinary student teams through **end-to-end ML project lifecycles**: problem framing, model selection, evaluation, and deployment on real industry datasets.
- Provide hands-on technical guidance in **Python, PyTorch, and applied ML**, emphasizing reproducibility, experimentation rigor, and ethical AI practices.

Johns Hopkins University — Baltimore, MD

Graduate Teaching Assistant – Artificial Intelligence — *Aug 2025 – May 2026*

- Support 60+ graduate students on **reasoning-based agents, probabilistic AI, and ML workflow debugging** through weekly office hours, structured feedback, and supplementary materials on Bayesian inference and search algorithms.

Graduate Teaching Assistant – Operating Systems — *Jan 2025 – May 2025*

- Guided students through **concurrency, synchronization, scheduling, and kernel-level debugging** in Pintos, strengthening systems-level understanding relevant to **high-performance serving infrastructure**.

Machine Learning Research Assistant — *Sept 2024 – Jan 2025*

- Developed **evaluation benchmarks for LLM-based table reasoning and complex database operations**, enabling systematic analysis of model accuracy, reasoning failures, and performance across structured-data tasks.

University of Essex — Colchester, UK

Artificial Intelligence Research Assistant — *Sept 2023 – Aug 2024*

- Built a **multi-modal sentiment analysis model** (PyTorch, TensorFlow, OpenAI) achieving **94% emotion classification accuracy**, contributing to **25% improvement** in early detection of Body Dysmorphic Disorder.
- Co-authored a **peer-reviewed publication** in *Human Behavior and Emerging Technologies* (Wiley, 2026) on a digital mental-health intervention; managed data pipelines and model versioning across a 3-person research team.

Salesforce — Remote

Software Engineer Intern — *May 2023 – Dec 2023*

- Integrated **real-time ML scoring** into lead routing workflows via Lightning Web Components and Apex, **reducing manual CRM processing by 30%** across the sales pipeline.
- Achieved **90% test coverage** on Apex classes; deployed metadata components via Salesforce DX and Git CI/CD, reducing release cycle friction.

Education

Johns Hopkins University — Baltimore, MD

Master of Science in Computer Science — *Aug 2024 – May 2026*

- Relevant Coursework: AI, Deep Learning, NLP, AI System Design, ML: Reinforcement Learning, Cloud Computing, ML: Learning Theory, CUDA Programming

Amity University — Noida, UP

BTech Hons. in Computer Science and Engineering (Specialization in AI/ML) — *Aug 2020 – May 2024*

- Relevant Coursework: Data Structures & Algorithms, Neural Networks, OS, Parallel Computing, Computer Architecture
- Study Abroad: Adelphi University (New York, Apr–Jun 2023) | University of London (London, Jan–Mar 2023)